

Lab 4: VibeCoding

Chat history · Space Invaders (SETAPESU26/46_space-invaders) · AI tool: Claude · 30 Sep 2026

Summary

Item	Outcome
Bug: can't shoot enemies	Fixed. Firing polled every frame (hold-to-fire) + text-input mode disabled; backup fire keys W / Up / Enter
Task 1: Collision detection	Done. Swept-rect collision; no removing from a list while iterating
Task 2: Game Over screen	Done. Overlay with final score; waits for input
Task 3: Replay option	Done. 1 = Easy, 2 = Medium, 3 = Hard, Q / Esc = Quit
Task 4: Sound feedback	Done. Generated sounds for fire, explosion, game over
Submission	Code, before/after videos and this PDF in Lab-4/ of the forked repo

Conversation

Prompts appear exactly as sent. Claude's replies are condensed; actions Claude took on the code are in *italics*.

PROMPT 1

[Attached: screenshot of the Lab 4 assignment + fixed_code_v2.zip]

Use Vibe Coding tools to fix the broken code and add features to the same...

https://github.com/SETAPESU26/46_space-invaders

so this is what my teacher gave as my lab-4 assignment... guide me step by step for the submission. so i am done till this, so yeah please correct whatever has to be corrected and i wanna tell u that m not able to shoot the enemies

CLAUDE

Claude unzipped the code, read every file, read the repo README (Tasks 1-4), and ran the game headlessly with simulated Space presses to test it.

The code already covers all four README tasks: swept-rect collision (Task 1), Game Over screen with final score (Task 2), replay with Easy/Medium/Hard (Task 3), generated sound effects (Task 4). In the test, bullets hit enemies and the score rose, so the logic was fine.

Why shooting failed: firing only happened on a single KEYDOWN event, so presses during the cooldown were dropped and holding Space did nothing; also Pygame 2's text-input mode can swallow Space on some keyboards/IMEs.

Fix: moved shooting into `handle_input()` using `pygame.key.get_pressed()` every frame (hold-to-fire, 12-frame cooldown) and added `pygame.key.stop_text_input()` in `main.py`. Re-tested: score rose while holding Space.

Also gave submission steps: replace files, record the after video, export chat, create Lab-4 folder, commit and push (no PR to SETAPESU26).

PROMPT 2

what am i supposed to do after downloading the zip file

CLAUDE

Step-by-step: extract the zip, copy main.py and game/ into the cloned repo (replace old files), run `python main.py`, click the window and test shooting, record the after video, put videos + code + chat PDF in a Lab-4 folder, then git add / commit / push.

PROMPT 3

m not able to shoot

CLAUDE

Claude added more robust input handling and diagnostics, then re-tested headlessly (score 11 in the test).

v3 changes: backup fire keys (Space / W / Up / Enter), firing on both key-tap and key-hold, a debug print of every key pygame receives, and a yellow 'v3' label on screen to confirm the new code is actually running.

Diagnosis guide: no 'v3' label means old code is running; label but no key prints means the window lacks focus; W works but Space doesn't means the keyboard layout is eating Space.

PROMPT 4

fine give me the commands after downloading the zip file

CLAUDE

Gave copy-paste commands for Windows (PowerShell) and Mac (Terminal) to unzip, copy files into the repo, install requirements and run.

PROMPT 5

[Pasted Mac terminal output: unzip could not find space_invaders_fixed_v3.zip; cd: no such file or directory: PATH/TO/46_space-invaders; cp failed; requirements.txt not found]

CLAUDE

Explained the two issues: the zip had a different name / was auto-extracted, and PATH/TO was a placeholder. The repo is at `~/Desktop/46_space-invaders`. Gave `ls ~/Downloads | grep -i space` to find the file and corrected copy commands.

PROMPT 6

i uncompressed it and saved it in downloads

CLAUDE

Gave a command block that finds the v3 code automatically and copies it in:

```
SRC=$(dirname "$(grep -rI "FIRE_KEYS" ~/Downloads --include=game_engine.py | head -n1)")
cp -f "$SRC/main.py" ~/Desktop/46_space-invaders/
cp -f "$SRC/game/*.py" ~/Desktop/46_space-invaders/game/
cd ~/Desktop/46_space-invaders
grep -c "FIRE_KEYS" game/game_engine.py
python main.py
```

PROMPT 7

fine everythings working now
wt to do later

CLAUDE

Remaining steps: remove the debug print and v3 label with sed, record the after video (Cmd+Shift+5), export the chat, create the Lab-4 folder, commit and push, verify on GitHub.

```
sed -i '' '/Key pressed:/d; /# Debug: shows every key/d' game/game_engine.py  
sed -i '' 's/v3 | Fire: Space \\/ W \\/ Up \\/ Enter/Fire: Space/' game/game_engine.py
```

PROMPT 8

first where to find the recorded video. i dont think anything changed in the repo cuz i just forked it and cloned it and i havent created the lab thingy in the repo, so guide me how to do all this

CLAUDE

Recordings save to the Desktop by default. The local repo did change (the new files were copied in) but isn't pushed yet; the fork is the student's own repo, so Lab-4 goes inside it. Steps: `git status` to confirm changes, create Lab-4/code, copy code + before/after videos + chat PDF, add a .gitignore for venv/, then commit and push (gh auth login if a password is requested).

PROMPT 9

[Attached: Finder screenshot of 46_space-invaders showing game, main.py, README.md, requirements.txt, venv]

what is this, and i couldnt get the chat history so can you make a pdf of the chat history. and those screen recordings are on the desktop

CLAUDE

Explained the folder contents and generated this PDF.

PROMPT 10

can you put the promp chat in a good way please

CLAUDE

Redesigned this PDF: summary table plus chat-style layout. Prompts kept word-for-word.